4.3.5 *Higher Order Symplectic Methods:* SHAKE *and* RATTLE

The projected symplectic constrained method (4.20)–(4.24) is only first order accurate. We forego providing a detailed proof of this fact, but note that it could be demonstrated using standard methods [164]. Note that (4.20)–(4.24) reduces to the symplectic Euler method in the absence of constraints, and the projection of the momenta would not alter this fact. There are several constraint-preserving, second-order alternatives which generalize the Störmer-Verlet scheme. One of these is the SHAKE method [322]. The original derivation of the SHAKE method began from the position-only, two-step form of the Störmer rule for $\ddot{\boldsymbol{q}} = \boldsymbol{F}(\boldsymbol{q})$

$$\boldsymbol{q}_{n+1} - 2\boldsymbol{q}_n + \boldsymbol{q}_{n-1} = h^2 \boldsymbol{M}^{-1} \boldsymbol{F}(\boldsymbol{q}_n).$$

In the presence of the constraint, we consider the scheme

$$\boldsymbol{q}_{n+1} - 2\boldsymbol{q}_n + \boldsymbol{q}_{n-1} = h^2 \boldsymbol{M}^{-1} \boldsymbol{F}(\boldsymbol{q}_n) - h^2 \boldsymbol{g}'(\boldsymbol{q}_n)^T \lambda_n, \tag{4.25}$$

$$\boldsymbol{g}(\boldsymbol{q}_{n+1}) = 0. \tag{4.26}$$

The symmetry suggests that this will be second order. In order to understand the symplectic property associated to this method, we need to define the updates for both positions and momenta. A natural choice is to consider the phase space formulation of the Störmer-Verlet method for $\dot{\boldsymbol{q}} = \boldsymbol{M}^{-1}\boldsymbol{p}, \dot{\boldsymbol{p}} = \boldsymbol{F}(\boldsymbol{q})$, then replace $\boldsymbol{F}$ by $\boldsymbol{F} - \boldsymbol{g}'(\boldsymbol{q})^T\lambda$, resulting in

$$\boldsymbol{q}_{n+1} = \boldsymbol{q}_n + h\boldsymbol{M}^{-1}\boldsymbol{p}_{n+1/2}, \tag{4.27}$$

$$\boldsymbol{p}_{n+1/2} = \boldsymbol{p}_n + (h/2)\boldsymbol{F}(\boldsymbol{q}_n) - (h/2)\boldsymbol{g}'(\boldsymbol{q}_n)^T\boldsymbol{\lambda}_n, \tag{4.28}$$

$$\boldsymbol{p}_{n+1} = \boldsymbol{p}_{n+1/2} + (h/2)\boldsymbol{F}(\boldsymbol{q}_{n+1}) - (h/2)\boldsymbol{g}'(\boldsymbol{q}_{n+1})^T\lambda_{n+1}. \tag{4.29}$$

It is possible to demonstrate (see exercises) that this method reduces to (4.25)–(4.26) by eliminating the momenta. However (4.27)–(4.29) does not give a map of the cotangent bundle, since $\boldsymbol{g}'(\boldsymbol{q}_{n+1})\boldsymbol{p}_{n+1} = 0$ will not be preserved, even if both the constraint and hidden constraint hold at step n.

A method that automatically preserves both the constraint and hidden constraint is the RATTLE method [12] which consists of the following steps:

$$\boldsymbol{q}_{n+1} = \boldsymbol{q}_n + h\boldsymbol{M}^{-1}\boldsymbol{p}_{n+1/2}, \tag{4.30}$$

$$\boldsymbol{p}_{n+1/2} = \boldsymbol{p}_n + (h/2)\boldsymbol{F}_n - (h/2)\boldsymbol{g}'(\boldsymbol{q}_n)^T\boldsymbol{\lambda}_n, \tag{4.31}$$

$$\boldsymbol{0} = \boldsymbol{g}(\boldsymbol{q}_{n+1}), \tag{4.32}$$

$$\boldsymbol{p}_{n+1} = \boldsymbol{p}_{n+1/2} + (h/2)\boldsymbol{F}_{n+1} - \boldsymbol{g}'(\boldsymbol{q}_{n+1})^T\boldsymbol{\mu}_{n+1}, \tag{4.33}$$

$$\boldsymbol{0} = \boldsymbol{g}'(\boldsymbol{q}_{n+1})\boldsymbol{M}^{-1}\boldsymbol{p}_{n+1}. \tag{4.34}$$